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Preliminary Insights Into the Relationship Between the Gut Microbiome and Host Genome in Posttraumatic Stress Disorder

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ABSTRACT

Posttraumatic stress disorder (PTSD) may develop following trauma exposure; however, not all trauma-exposed individuals develop PTSD, suggesting the presence of susceptibility and resilience factors. The gut microbiome and host genome, which are interconnected, have been implicated in the aetiology of PTSD. However, their interaction has yet to be investigated in a South African population. Using genome-wide genotype data and 16S rRNA (V4) gene amplicon sequencing data from 53 trauma-exposed controls and 74 PTSD cases, we observed no significant association between the host genome and summed abundance of *Mitsuokella, Odoribacter, Catenibacterium* and *Olsenella*, previously reported as associated with PTSD status in this cohort. However, *PROM2* rs2278067 T-allele was significantly positively associated with the summed relative abundance of these genera, but only in individuals with PTSD and not trauma-exposed controls ($p < 0.014$). Polygenic risk scores generated using genome-wide association study summary statistics from the PGC-PTSD Overall Freeze 2 were not predictive of gut microbial composition in this cohort. These preliminary results suggest a potential role for the interaction between genetic variation and gut microbial composition in the context of PTSD, underscoring the need for further investigation.

1 | Introduction

Posttraumatic stress disorder (PTSD) develops following exposure to a traumatic event [1]. Globally, the lifetime prevalence of PTSD is 5.6% among trauma-exposed individuals [2]. In South Africa, one study reported that 73.8% of individuals have been exposed to at least one lifetime potentially traumatic event, while the lifetime prevalence of PTSD in South Africa is estimated at 2.3%, although authors warn this is likely an underestimation [3]. PTSD has a multifactorial aetiology with demographic and environmental contributions, such as age, sex, marital status, employment status, education level and household income [2].

There is growing interest in the role of the gut microbiome in neuropsychiatric disorders, with accumulating evidence suggesting that gut microbial composition may be associated with the aetiology of several psychiatric disorders, including major depressive disorder [4], bipolar disorder [5], schizophrenia [5] and PTSD [6]. Gut microbial composition can affect brain functioning via the microbiota-gut-brain axis, a bidirectional axis of communication between the gut, the gut microbiome and the central nervous system [7]. Communication within the axis occurs via multiple complementary routes, including the autonomic nervous system (including the vagus nerve), the enteric nervous system, the immune system, enteroendocrine signalling, neurotransmitters, branched chain amino acids, bile moieties, short-chain fatty acids, spinal mechanisms and the hypothalamic–pituitary–adrenal axis (see review by Cryan et al. [8]). However, the precise mechanisms underpinning the interaction between the gut microbiome and psychiatric disorders are understudied and only limitedly understood.

The use of murine models has generated a substantial amount of evidence to support the role of the gut microbiome in stress and trauma response [9, 10], and, recently, the contribution of the gut microbiome to PTSD development has been observed in humans [6, 11]. In South Africa, only three studies have been conducted on the gut microbiome and PTSD to date, all of which identified differences in gut microbial composition based on PTSD status [12–14]. The first, by Hemmings et al. [12], reported decreased total abundance of the phyla Actinobacteria, Lentisphaerae and Verrucomicrobia in individuals with PTSD. The second study represented a larger cohort, which included some of the same participants from the first pilot study, and reported an increased abundance of a consortium of four genera, that is, a combination of *Mitsuokella*, *Odoribacter*, *Catenibacterium* and *Olsenella*, in individuals with PTSD and in individuals with more severe childhood trauma [13]. The third study, in a different South African cohort, found the abundances of *Catenibacterium*, *Collinsella* and *Holdemanella* to each be significantly positively associated with the severity of PTSD symptoms [14].

Genetics is also a contributory factor in the aetiology of PTSD. Genome-wide association studies (GWAS) and polygenic risk scores (PRS) have allowed identification of genetic risk variants and prediction of phenotypes associated with PTSD, respectively. The largest PTSD GWAS meta-analysis to date identified 95 risk loci, 80 of which were new, and several of which implicated immune processes [15]. Swart and co-workers reported the first GWAS and PRS analysis of PTSD in a South African population using a superset of the data in the current study [16]. They found no genome-wide significance of SNPs associated with PTSD, but they did find variants of *PARK2* that may be associated with the development of PTSD and metabolic syndrome. Despite the modest sample (PTSD participants $n = 260$; trauma-exposed controls $n = 343$), their findings regarding *PARK2* were in agreement with those reported in the largest PTSD GWAS at the time [17]. The authors suggested that examining the genomic data in conjunction with other omics data would increase the discovery of new insights [16]. In the last five years, there has been a growing interest in performing microbiome and GWAS analysis to elucidate the heritable traits of the microbiome in the context of psychiatric disorders.

Host–microbiome genome-wide association studies (mGWAS) have identified many genetic variants associated with taxonomic abundance [18–22]. This host genome-gut microbiome relationship has also been shown to hold true in the context of neurological and psychiatric disorders [11, 23]. For instance, one recent study reported a link between the abundance of 10 genera and host genetic factors that may promote the development of Alzheimer's disease [24]. Another recent mGWAS reported the composition of the nasal microbiome to be associated with 63 genome-wide significant loci, some of which are implicated in neuropsychiatric disorders, such as schizophrenia [25]. A theme that somewhat consistently emerges throughout mGWAS is the interaction between the gut microbiome, host genome and immune system [26, 27].

Given the imperative need for inclusion of diverse populations in mGWAS [28], we aim to investigate whether the gut microbiome and host genome are associated in the context of PTSD among South African individuals. The current study is based on data previously generated in our group. We used gut microbial data from Malan-Müller et al. [13] to investigate whether the summed abundance of *Mitsuokella*, *Odoribacter*, *Catenibacterium* and *Olsenella* was associated with PTSD, within a smaller sample subset of the original Malan-Müller et al. [13] paper. We used genotype data from Swart et al. [16] to investigate whether any genetic variants were associated with PTSD in our subset, and could predict PTSD status and the summed abundance of the four genera in this cohort [16]. However, to account for the limited sample size in the current study, we only investigated SNPs previously associated with gut microbial composition [29].

2 | Results

2.1 | Study Population Description

Participants with PTSD were significantly younger than trauma-exposed controls (TEC, Wilcoxon, $p < 0.05$). Sex did not significantly differ between PTSD cases and TEC groups (χ^2, $p > 0.05$). Individuals with PTSD had higher Clinician-Administered Posttraumatic Stress Disorder Scale for DSM-5 (CAPS-5) total scores (Wilcoxon, $p < 0.05$) and childhood trauma questionnaire (CTQ) total scores (Wilcoxon, $p < 0.05$) compared to TEC (Table 1).

The summed relative abundance of the four genera was higher in individuals with PTSD compared to TEC (Wilcoxon,

$p = 0.001$) and was positively correlated with CAPS-5 total scores (Spearman $r_s = 0.28$; $p = 0.001$), although not when stratified into PTSD cases (Spearman $r_s = 0.124$; $p = 0.293$) and controls (Spearman $r_s = -0.049$; $p = 0.725$) (Figure 1A,B). The summed relative abundance of these genera was not significantly associated with CTQ total scores (Spearman $r_s = 0.14$; $p = 0.09$) (Figure 1C).

2.2 | Host Genome-Microbiome Analysis

Ten out of a total of 143 SNPs were associated with the sum of the four genus-level taxa ($p < 0.05$) (Figure 2): rs2710119 (contactin-associated protein-like 2, *CNTNAP2*), rs2278067 (prominin2, *PROM2*), rs61906997 (intergenic), rs3775467 (multimerin, *MMRN1*), rs13249293 (intergenic), rs2630788 (zinc finger protein 385D, *ZNF385D*), rs986417 (intergenic), rs115795847 (intergenic), rs11236806 (long intergenic non-protein coding RNA 2757, *LINC02757*) and rs17806643 (intergenic). However, these results were not significant following Bonferroni correction for multiple testing ($p > 0.0003$) (Figure S1A). Summary statistics are described in Table S1.

The relationship between genotypes with a p-value < 0.01 (rs2710119 and rs2278067), PTSD case/control status and the summed relative abundance of the four genus-level taxa is depicted in Figure 3. The SNP with the lowest p-value, rs2710119 (*CNTNAP2*), did not interact with PTSD status to influence the sum of the four genus-level taxa ($p > 0.05$, Figure 3A). However, the summed relative abundance of the four genus-level taxa increased with every copy of the rs2278067 (*PROM2*) T allele in PTSD cases but not in TEC ($\beta = 1.72$, SE $= 0.69$, $t = 2.49$, $p_{\text{interaction}} = 0.0138$, $F_{(9,117)} = 3.73$, $p_{\text{model}} = 0.0004$, Figure 3B).

TABLE 1 | Study cohort characteristics.

	PTSD ($n = 74$)	TEC ($n = 53$)	Test	p
Age, median (IQR)	42 (20)	51 (14)	$W = 1335$	0.0022*
Sex, n female (%)	59 (79.7)	43 (81.1)	$\chi^2 = 1.27 \times 10^{-31}$	1
CAPS-5 total score, median (IQR)	36.5 (11.8)	2 (7)	$W = 3922$	< 0.0001*
CTQ, median (IQR)	49 (33)	35 (19)	$W = 2755.5$	0.0001*

Abbreviations: CAPS-5, Clinician-Administered Posttraumatic Stress Disorder Scale for DSM-5; CTQ, Childhood Trauma Questionnaire; IQR, interquartile range (IQR = Q3 − Q1); n, number; TEC, trauma-exposed controls; W, Wilcoxon test; χ^2, Chi-squared test.
*Significance ($p < 0.05$).

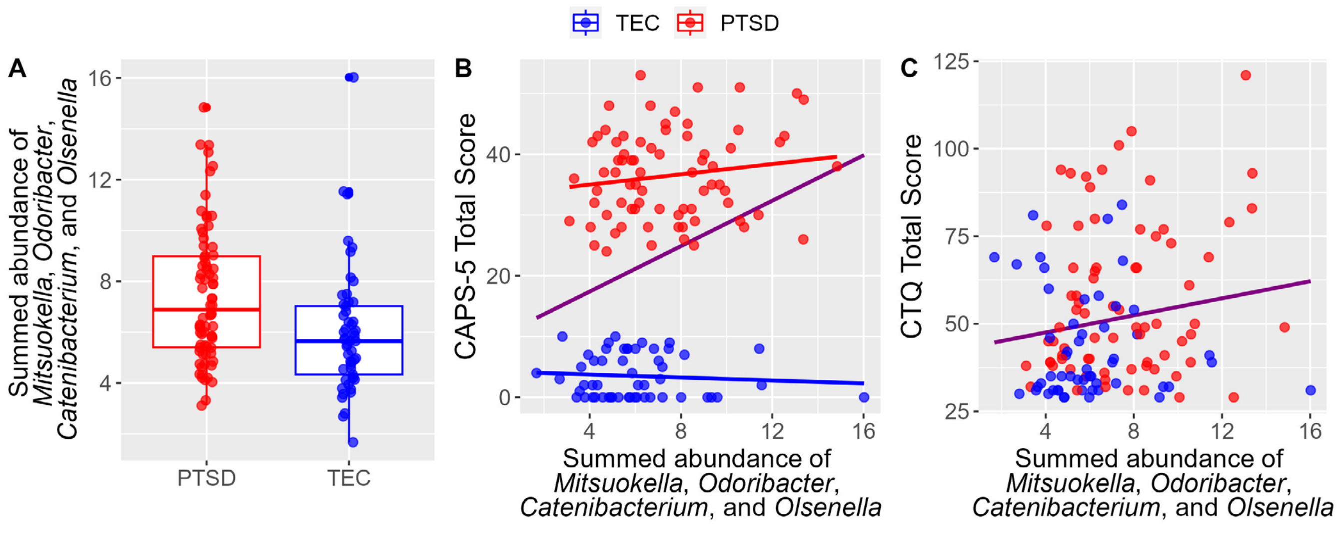

FIGURE 1 | The relationship between the summed relative abundance of four genus-level taxa and PTSD case/control status, CAPS-5 total scores and CTQ scores in $N = 127$ participants (PTSD = 74, TEC = 53). This figure is comparable to Figure 3 of Malan-Müller et al. [13] (Figure S2; $N = 137$), although only a subset ($N = 127$) with both microbiome data and host genome data was used in the current analysis. (A) The summed relative abundance of the four genera was higher in individuals with PTSD compared to TEC (Wilcoxon, $p = 0.001$) and (B) was positively correlated with the overall CAPS-5 total score (Spearman $r_s = 0.28$; $p = 0.001$ [purple]), although not when stratified into PTSD cases (Spearman $r_s = 0.124$; $p = 0.293$ [red]) and controls (Spearman $r_s = -0.049$; $p = 0.725$ [blue]). (C) The summed relative abundance was not significantly correlated with CTQ score (Spearman $r_s = 0.14$; $p = 0.09$). CAPS-5, Clinician-Administered Posttraumatic Stress Disorder Scale for DSM-5; CTQ, Childhood Trauma Questionnaire; IQR, interquartile range; TEC, trauma-exposed control.

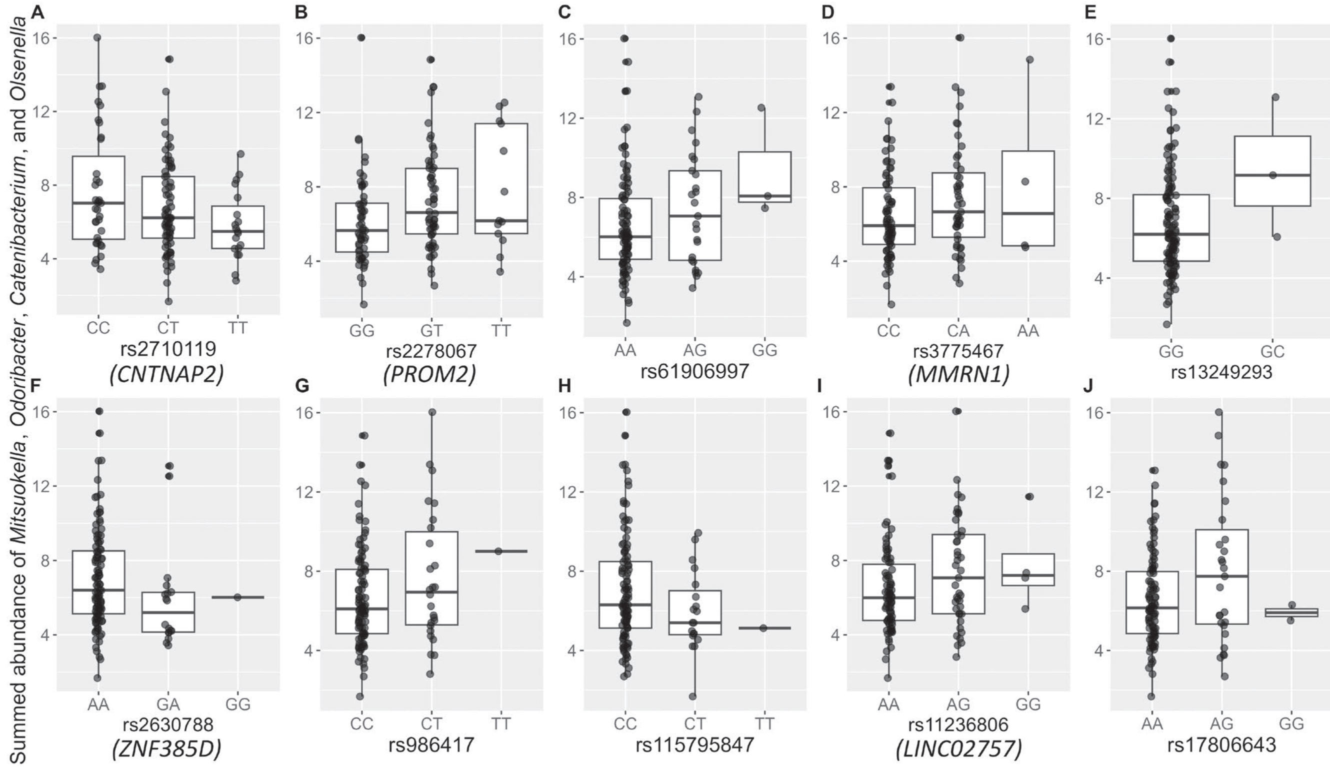

FIGURE 2 | The minor allele of 10 SNPs was associated with the total summed relative abundance of four genus-level taxa ($p < 0.05$). (A) rs2710119 (*CNTNAP2*) (CC = 35, CT = 73, TT = 19). (B) rs2278067 (*PROM2*) (GG = 56, GT = 58, TT = 13). (C) rs61906997 (AA = 99, AG = 25, GG = 3). (D) rs3775467 (*MMRN1*) (CC = 76, CA = 47, AA = 4). (E) rs13249293 (GG = 124, GC = 3). (F) rs2630788 (*ZNF385D*) (AA = 108, GA = 18, GG = 1). (G) rs986417 (CC = 100, CT = 26, TT = 1). (H) rs115795847 (CC = 107, CT = 19, TT = 1). (I) rs11236806 (*LINC02757*) (AA = 82, AG = 41, GG = 4). (J) rs17806643 (AA = 98, AG = 27, GG = 2). Significance was not retained following Bonferroni correction for multiple testing ($p > 0.0003$). Data are represented by box plots overlayed by individual data points. SNPs without names are unmapped.

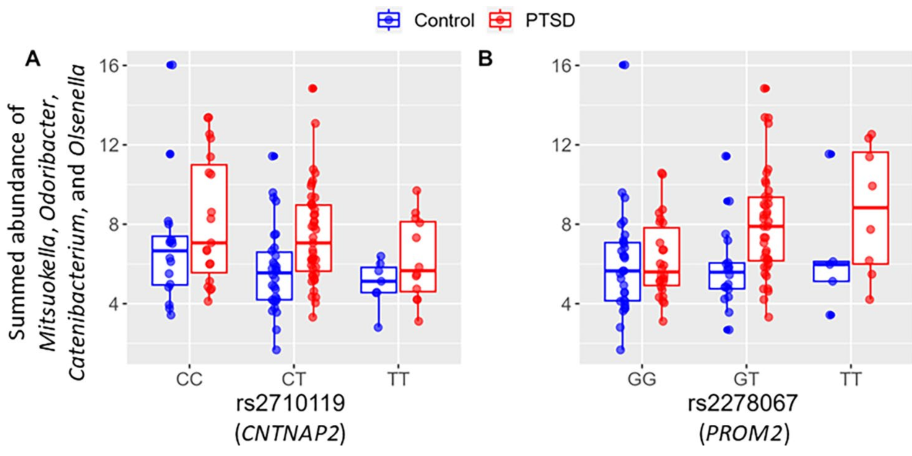

FIGURE 3 | *PROM2* rs2278067, but not *CNTNAP2* rs2710119, interacted with PTSD case/control status to influence the total relative abundance of four genus-level taxa. (A) rs2710119 (*CNTNAP2*) did not interact with PTSD status to influence the sum of the four genus-level taxa ($p > 0.05$) (Control [blue], CC = 16, CT = 30, TT = 7; PTSD [red], CC = 19, CT = 43, TT = 12). (B) The summed relative abundance of the four genus-level taxa increased with every copy of the rs2278067 (*PROM2*) T allele in PTSD cases but not in TEC ($\beta = 1.72$, SE = 0.69, $t = 2.49$, $p_{\text{interaction}} = 0.0138$, $F_{(9,117)} = 3.73$, $p_{\text{model}} = 0.0004$) (Control [blue], GG = 30, GT = 18, TT = 5; PTSD [red], GG = 26, GT = 40, TT = 8). Data are represented by box plots overlayed by individual data points.

2.3 | Polygenic Risk Score Analysis

The PTSD-PRS calculated in this subset of individuals ($N = 127$) using the PGC-PTSD Freeze 2 GWAS overall summary statistics was not significantly associated with PTSD case/control status nor the summed relative abundance of the four genus-level taxa (Figure 4). Regression analysis obtained the lowest p-value at a p-value threshold of 0.05 (PTSD-PRS ~ case/control status,

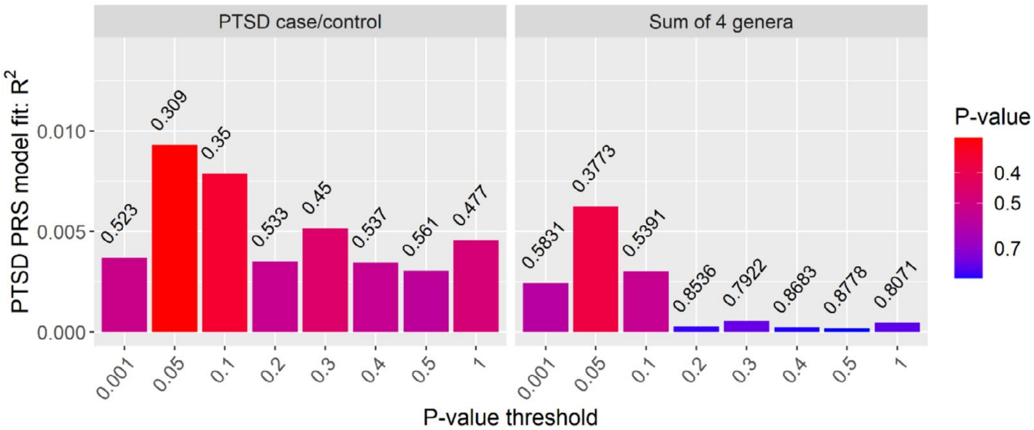

FIGURE 4 | The PRS for PTSD was not associated with PTSD case/control status or the relative abundance sum of four genus-level taxa. PTSD-PRS ~ case/control status, $p = 0.309$ (left-hand panel); PTSD-PRS ~ sum of four genera, $p = 0.377$ (right-hand panel).

$p = 0.309$; PTSD-PRS ~ sum of four genera, $p = 0.377$) explaining less than 1% of the variation (not significant).

3 | Discussion

We investigated the association between the gut microbiome and host genome in the context of PTSD in a South African sample by examining the relative abundance of a consortium of four bacteria found to be associated with PTSD in Malan-Müller et al. [13], with the current study using a subset of the participants from Malan-Müller et al. [13] that had both microbiome and host genome data. From a selection of SNPs previously found to be associated with gut microbial composition [29], we found 10 SNPs to be associated with the summed relative abundance of these 4 genera. However, none of these associations survived correction for multiple testing.

Interestingly, the *PROM2* rs2278067 T-allele was significantly positively associated with the summed relative abundance of *Mitsuokella*, *Odoribacter*, *Catenibacterium* and *Olsenella* in individuals with PTSD, but not in TEC. *PROM2* encodes a polytopic membrane protein that facilitates resistance to ferroptosis, a form of cell death related to iron-mediated oxidative damage to cell membranes [30]. It does this by promoting the export of iron from the cell via the formation of ferritin-containing multivesicular bodies and exosomes [31]. Ferroptosis has been linked to neurodevelopment, as well as neurological (e.g., amyotrophic lateral sclerosis [32]), neurodegenerative (e.g., Alzheimer's disease and Parkinson's disease [33]) and psychiatric disorders (e.g., PTSD [34]). Ferroptosis has also been suggested to alter immune status in the context of PTSD via the release of proinflammatory factors [34]. While ferroptosis and inflammation are important factors to investigate in the relationship between PTSD, the gut microbiome and the host genome, they are beyond the scope of the current exploratory study, as biomarkers representative of these processes are not currently available in this cohort.

The PRS generated using the PGC-PTSD Freeze 2 as the discovery dataset did not explain a significant proportion of the variance regarding PTSD case/control status nor the summed abundance of the four genera. However, results from our PRS analysis need to be interpreted with caution due to the paucity of available GWAS summary statistics from large non-European or admixed ancestry populations that could be used as discovery data at the time of these analyses.

Although analysis of the larger dataset (from [13]) found a significant association between childhood trauma and the summed abundance of the four genera, this association was not significant in the current study. The smaller sample size in the current study may have affected the power to detect an association since the overall participant characteristics (median age, % females, median CAPS-5 score, median CTQ score) remained proportionally similar between this study and that of Malan-Müller et al. [13].

While this exploratory study demonstrated that the gut microbiota and the host genome are linked in the context of PTSD, some limitations should be mentioned. First, although we attempted to enhance the power to detect association by only investigating the total abundance of four bacteria and certain SNPs previously associated with gut microbial composition, the limited sample size meant that the GWAS and the PRS analyses were likely underpowered. Second, the techniques used herein to generate PRS are potentially not well-suited to incorporate the complex genetic makeup associated with South African populations, given that most of the samples for the discovery dataset comprise samples with European ancestry and the South African samples in this study comprise a much richer five-way admixed ancestry. Regarding the microbiome analyses, sequencing was only performed on the 16S ribosomal RNA (rRNA) gene V4 region, which limited taxonomic resolution to the genus level. This may have resulted in the loss of valuable insights as various species and strains within the same genus may have different byproducts and functions. Additionally, the use of hypervariable sequencing data limited the ability to conduct in-depth, accurate inference of the functional pathways of the microbes and their products that may interact with the host genome.

Despite these limitations, this study provides valuable preliminary insights into the relationship between the host genome and the gut microbiome in the context of PTSD—a disorder that is substantially understudied relative to other psychiatric disorders. Further, this study has been conducted in a South African sample, thereby contributing to the limited knowledge base of

African and South African gut microbiome studies. It is imperative that future studies replicate and validate these findings in a larger cohort. Additionally, subsequent investigations should explore levels of inflammatory mediators associated with PTSD to elucidate the underlying mechanisms of interaction and may gain more insight by stratifying groups based on PTSD symptom severity and duration. Understanding the relationship between the gut microbiome and host genome in the context of PTSD may aid in addressing the missing heritability problem in PTSD [35], and controlling for environmental factors when conducting GWAS [36].

4 | Material and Methods

4.1 | Study Participants

Participants were recruited as part of the parent study, Shared Roots of Neuropsychiatric Disorders and Cardiovascular Disease Project (Shared Roots; HREC no. N13/08/115; Department of Psychiatry, Faculty of Medicine and Health Sciences, Stellenbosch University) in Cape Town, South Africa. This study included 74 PTSD cases and 53 TEC for whom both microbiome and genome-wide genotype data were available from the Shared Roots project.

Participants were included if they were willing and able to provide written informed consent, 18 years of age or older, able to read and write Afrikaans or English, and not pregnant. Participants were excluded if they had a diagnosis of bipolar or psychotic disorders, alcohol or drug use disorders, or neurological disorders (Mini International Neuropsychiatric Interview version 6.0, MINI; [37]). Participants were excluded from gut microbiome analysis if they had experienced diarrhoea within the past week, had been on antibiotics within 4 weeks before stool sampling, or had a diagnosis of inflammatory bowel disease, irritable bowel syndrome, or coeliac disease (all criteria self-reported by the participant).

PTSD symptoms over the previous month were assessed using CAPS-5, where participants with a severity score greater than 23 were classified as PTSD cases [38]. Control participants had a CAPS-5 score less than 11 and were matched on ethnicity, age and sex (ascertained using a demographic questionnaire) and were trauma-exposed based on the DSM-5 criteria (the experience of actual or threatened death, serious injury, or sexual violence). Childhood maltreatment was assessed using the CTQ total scores [39].

4.2 | Microbiome Data

This study uses the Shared Roots dataset that was previously analysed by Malan-Müller et al. [13] containing 16S rRNA (V4) gene amplicon data from 79 individuals with PTSD and 58 TECs [13]. Briefly, participants self-collected stool samples during the week of clinical assessments using PSP stool collection tubes containing DNA stabilizing buffer (STRATEC Molecular, Birkenfeld, Germany). Microbial DNA was extracted using the PSP Spin Stool DNA Plus Kit (STRATEC Molecular, Birkenfeld, Germany) according to the manufacturer's instructions. Amplicons of the 16S rRNA gene were generated using the 515-bp forward (5′-*GTGCCAGCMGCCGCGGTAA*-3′) and 806-bp reverse (5′-*GGACTACHVGGGTWTCTAAT*-3′) primer pair [40] on the Illumina MiSeq platform (300 bp paired-end sequencing). Raw reads were processed using *DADA2* (version 1.12.1) [41]. The Ribosomal Database Project Classifier (Train Set 16, release 11.5; [42]) was used for the taxonomic binning of classified sequences. A feature table of 2593 unique amplicon sequence variants with an average read length of 253 bp in 137 samples was used (the minimum number of reads per sample was 19,388 following pre-processing).

To evaluate microbiome community variation, taxa were agglomerated to genus level and abundance matrices were centre log-ratio (clr)-transformed with the *codaSeq.clr* function using the CoDaSeq package in R [43]. Filtering was performed to retain taxa that were observed in at least 15% of participants (to eliminate taxa with very low abundance). Random forest, a supervised machine learning algorithm, was then used to evaluate if features of the gut microbiome can predict PTSD status. The R package *randomForests* [44] was used on 95 genera agglomerated from the filtered ASV table. This method implements Breiman's random forest algorithm for classification [44]. The previous findings by Malan-Müller et al. [13] showed that the summed relative abundance of four genera, namely *Mitsuokella*, *Odoribacter*, *Catenibacterium* and *Olsenella*, was higher in individuals with PTSD compared to TEC and positively correlated with CAPS-5 score (Spearman $r_s = 0.25$; $p = 0.003$) and CTQ score (Spearman $r_s = 0.17$; $p = 0.05$) [13] (Figure S2 and Table S2). In the current study, the summed relative abundance of these four genera was used as the outcome variable in linear regression models to investigate the relationship between the gut microbiome and host genetic variation in our subset of the original cohorts.

4.3 | Genotype Data

This sub-study of Shared Roots used genotype data previously generated [16]. DNA from all participants was extracted from blood samples using the Gentra Puregene Blood Kit (Qiagen), according to the manufacturer's protocol, in the Neuropsychiatric Genetics Laboratory at Stellenbosch University, South Africa. Genome-wide genotyping was conducted using the Multi-ethnic Global Array (MEGA, Illumina) and the Infinium Global Screening Array (GSA, Illumina). Quality control and imputation procedures are described in detail in Swart et al. [16]. Briefly, variants were included if they had a minor allele frequency (MAF) > 1%, a missingness rate of < 3%, had the same call rate between PTSD cases and TEC, and were in Hardy–Weinberg Equilibrium. Imputation was conducted via the Sanger Imputation Server and the African Genome Resource (https://www.apcdr.org/), which has been shown to be the best publicly available reference panel for this population group [45]. Following imputation, the genotyping data consisted of SNPs with an INFO Score > 0.8, a MAF > 1% and a missingness rate < 3%.

PLINK 1.9 [46] was used to extract ("–*extract*") 180 SNPs, previously found to be associated with the gut microbiome [29], from the quality-controlled, imputed genome-wide genotype data [16], available for 127 of the 137 participants used in Malan-Müller et al. [13]. The overlapping 143 SNPs were quality-controlled to ensure a MAF > 1% and a SNP missingness rate of < 3%. SMARTPCA [47, 48] was used to perform principal component analysis using the 143 SNPs and 127 participants (74 PTSD, 53 TEC). PTSD cases and TEC were evenly distributed across the principal components (Figure S3). The first five principal components were used in the association models to account for population stratification and to manage the complexity of multiple variables contributing to the outcome.

4.4 | Host Genome-Microbiome Association Analysis

Linear regression models were used to test for an association between 143 SNPs and the summed relative abundance of four genus-level taxa in PLINK 1.9. The first five principal components, genotyping batch and PTSD case/control status were used as covariates (Model 1). SNPs were considered significantly associated with the summed relative abundance after Bonferroni correction for multiple testing ($p < 0.0003$). A p-value between 0.0003 and 0.05 was assigned as a probable association. Similarly, interaction effects between genotype and PTSD case/control status on summed relative abundance were investigated using SNPs that demonstrated a probable association with summed relative abundance (Model 2).

Model 1: Sum of 4 genus-level taxa ~ genotype + PC1 + PC2 + PC3 + PC4 + PC5 + batch + case/control

Model 2: Sum of 4 genus-level taxa ~ genotype * case/control + PC1 + PC2 + PC3 + PC4 + PC5 + batch

4.5 | Polygenic Risk Scores

The polygenic risk for PTSD was previously calculated in Shared Roots participants ($N = 603$, [16]) using the PGC-PTSD Freeze 2 GWAS summary statistics [17] and PRSice [49]. The PRS calculated at each of the p-value thresholds (0.01, 0.05, 0.1, 0.2, 0.3, 0.4, 0.5) using the PGC-PTSD Freeze 2 GWAS overall (ALL) summary statistics for 127 participants were regressed onto the first five principal components. The phenotype of interest (PTSD case/control status and summed relative abundance of the four genera) was then regressed on the standardized residuals. The variance explained by the regression model was denoted by Nagelkerke's pseudo-R^2 value if the outcome variable was binary (PTSD case/control status) or R^2 if the outcome variable was continuous (summed relative abundance of the four genera).

4.6 | Statistical Analysis

Statistical analysis was conducted in R 4.1.2 or higher [50]. The package *ggplot2* [51] was used to generate figures. Shapiro Wilk's test was used to assess the distribution of continuous data (age, CAPS-5 total score, CTQ score, and summed relative abundance of the four genera). Differences in non-parametric data (age, CAPS-5 total score, CTQ score, and summed relative abundance of the four genera) between individuals with PTSD and TEC were analysed using a Wilcoxon rank sum test and were represented by the median and interquartile range. Differences in categorical variables (sex) between PTSD cases and TEC were analysed using a Chi-squared (χ^2) test. Spearman's correlation was used to test for correlations between summed relative abundance of the four genera and CAPS-5 and CTQ scores.

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Ethics Statement

Ethics approval was obtained from the Health Research Ethics Committee at Stellenbosch University (N13/08/115) and written informed consent was obtained from all participants.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The datasets presented in this article are not readily available due to ethical and legal restrictions. Requests to access the datasets should be directed to S.M.J.H. (smjh@sun.ac.za). The authors are open to collaborating and sharing data within the limits of ethical review restrictions and data transfer policies of Stellenbosch University.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.